

ASSIGNMENT ON CSE-420

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Answer to the Question (CSDT Rules)

Production	Semantic Rules
$S \rightarrow id$	$\{ \text{sym} = \text{table.get}(id.lexeme); \text{if } (\text{sym} \neq \text{NULL})$ $S.type = \text{sym.type}; \text{else } S.type = \text{error}; \}$
$S \rightarrow S_1[E]$	$\{ \text{if } (S_1.type == \text{error} \text{ or } E.type == \text{error}) S.type = \text{error};$ $\text{else if } (E.type \neq \text{int}) S.type = \text{error}; \text{else}$ $S.type = \text{element}(S_1.type); \}$
$E \rightarrow \text{num}$	$\{ \text{if } (\text{is_integer}(\text{num.value})) E.type = \text{int};$ $\text{else } E.type = \text{float}; \}$
$E \rightarrow S$	$\{ E.type = S.type; \}$
$E \rightarrow E_1 + E_2$	$\{ \text{if } (E_1.type == \text{error} \text{ or } E_2.type == \text{error})$ $E.type = \text{error}; \text{else if } (E_1.type == \text{int} \text{ and}$ $E_2.type == \text{int}) E.type = \text{int}; \text{else if } (E_1.type == \text{float}$ $\text{or } E_2.type == \text{float}) E.type = \text{float}; \text{else } E.type = \text{error}; \}$
$E \rightarrow E_1 * E_2$	$\{ \text{if } (E_1.type = \text{int} \text{ and } E_2.type == \text{int}) E.type = \text{int};$ $\text{else } E.type = \text{error}; \}$
$E \rightarrow (E_1)$	$\{ E.type = E_1.type; \}$
$P \rightarrow S = E$	$\{ \text{if } (S.type == \text{error} \text{ or } E.type == \text{error}) P.type =$ $\text{error}; \text{else if } (S.type == E.type)$ $P.type = \text{void}; \text{else if } (S.type == \text{float}$ $\text{and } E.type == \text{int}); P.type = \text{void};$ $\text{else } P.type == \text{error}; \}$